DO NOW — RECALL

Lossy and lossless compression

Name: _____ Class: _____ Date: _____

1. What can a firewall block?

2. What is non-volatile storage?

3. What is a bitwise mask?

4. Apply bitwise AND: $11011010_2 \wedge 00001111_2$. Give 8-bit answer and hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Traffic failing its security rules.

2. Model answer

Storage retaining content without power.

3. Model answer

Bit pattern applied with bitwise operators to select/change bits.

4. Model answer

$00001010_2 = 0A_{16}$ (last four bits 1010 retained).

Vocabulary — know these terms

lossless — Reconstructs original data exactly.

lossy — Permanently removes some information.

compression ratio — Original size compared with compressed size.

redundancy — Repeated/predictable information.

decompression — Recover encoded content.

NEW LEARNING

Read, annotate and explain

Understand

Lossless compression represents data more efficiently without discarding any information and reconstructs the original exactly.

Apply

Lossy methods intentionally discard information; a decoded image/audio can be perceptually similar but not bit-for-bit identical.

Evaluate

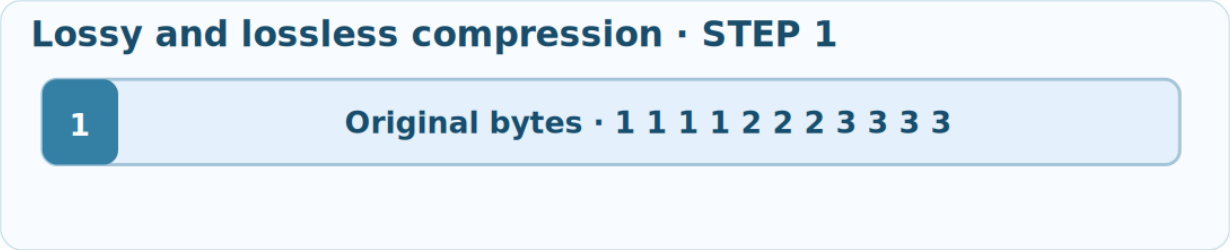
Choose lossless for text, programs, records or medical numeric originals; lossy can be appropriate for delivery copies of photos/video/audio with acceptable quality reduction.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

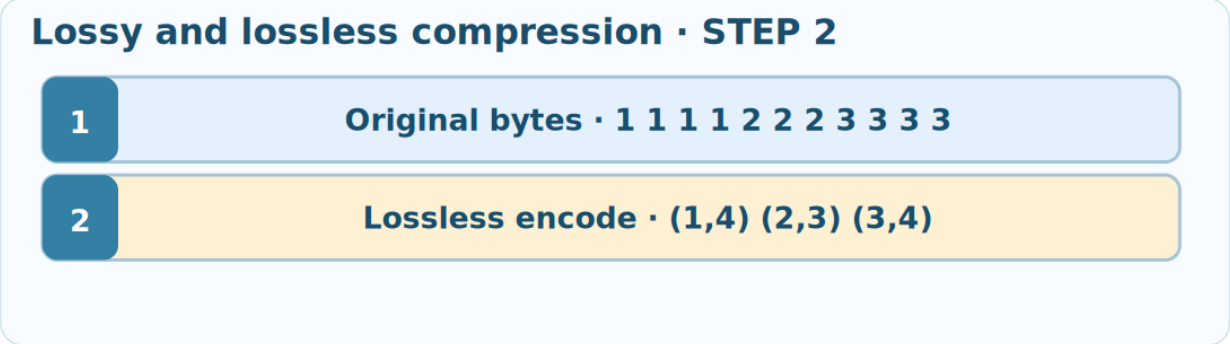
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Lossy and lossless compression · STEP 1: Original bytes · 1 1 1 1 2 2 2 3 3 3 3

New Learning

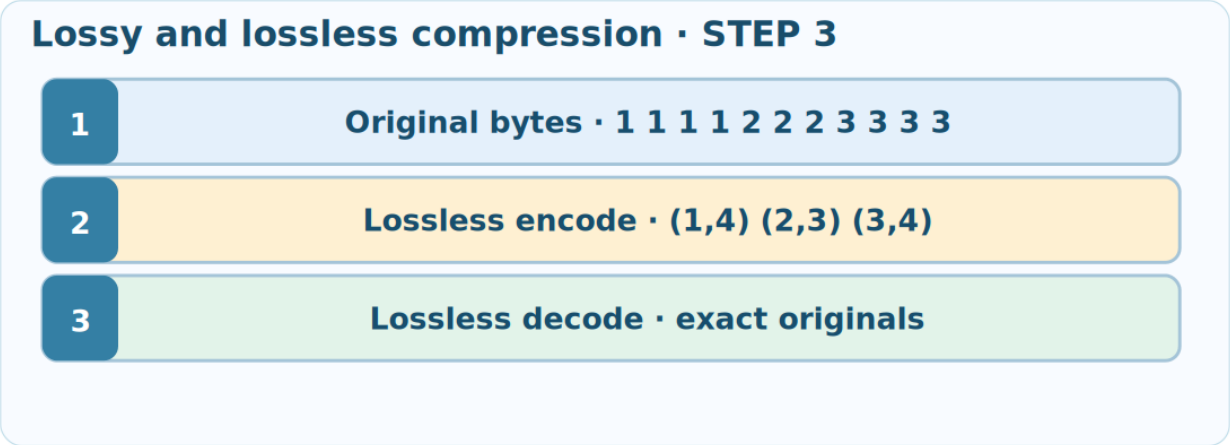


Lossy and lossless compression · STEP 2: Original bytes · 1 1 1 1 2 2 2 3 3 3 3; Lossless encode · (1,4) (2,3) (3,4)

New Learning · original lesson diagrams

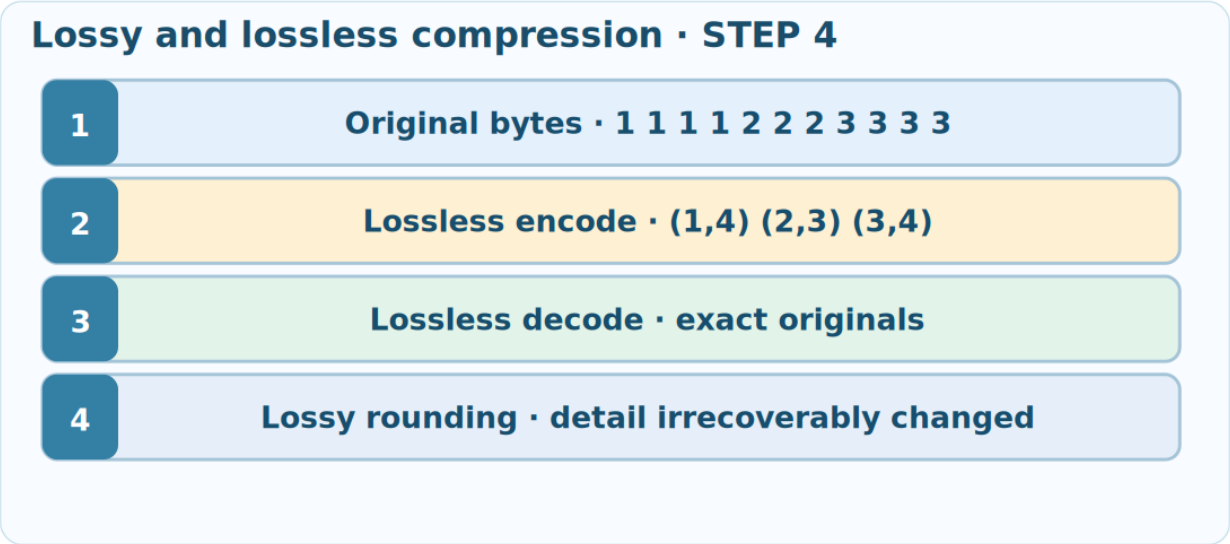
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Lossy and lossless compression · STEP 3: Original bytes · 1 1 1 1 2 2 2 3 3 3 3; Lossless encode · (1,4) (2,3) (3,4); Lossless decode · exact originals

New Learning



Lossy and lossless compression · STEP 4: Original bytes · 1 1 1 1 2 2 2 3 3 3 3; Lossless encode · (1,4) (2,3) (3,4); Lossless decode · exact originals; Lossy rounding · detail irrecoverably changed

I DO — WORKED EXAMPLE

Follow the worked method

A database contains precise 7-digit identifiers. Is lossy compression suitable?

Predict first

An altered digit can identify a different record and corrupt its meaning.

Model and check

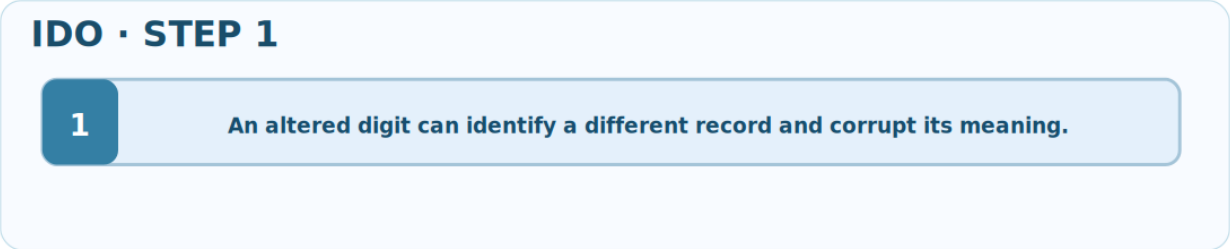
Lossless encoding is essential; compression ratio depends on redundancy in the source.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A database contains precise 7-digit identifiers. Is lossy compression suitable?

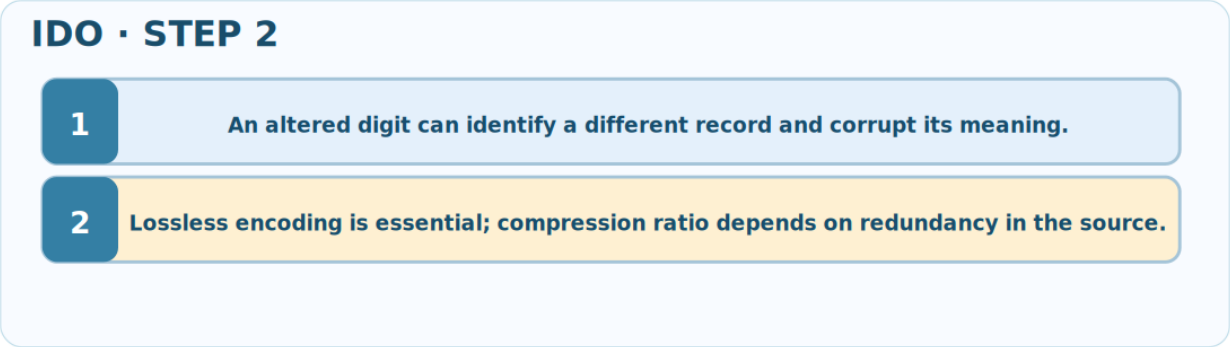
An altered digit can identify a different record and corrupt its meaning.



IDO · STEP 1: An altered digit can identify a different record and corrupt its meaning.

A database contains precise 7-digit identifiers. Is lossy compression suitable?

An altered digit can identify a different record and corrupt its meaning.



IDO · STEP 2: An altered digit can identify a different record and corrupt its meaning.; Lossless encoding is essential; compression ratio depends on redundancy in the source.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A web page uses 4K source photographs but displays small thumbnails. Evaluate lossy compression.

A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.



WEDO · STEP 1: A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define lossy and lossless compression; state one suitable use of each. [6]

Question 2

Given sequence A A A B B A, write RLE run/value pairs and show how to decode. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A school distributes music and exam marks electronically. Recommend encoding and justify trade-offs. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Lossy compression is suitable for any data if it produces the smallest file.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

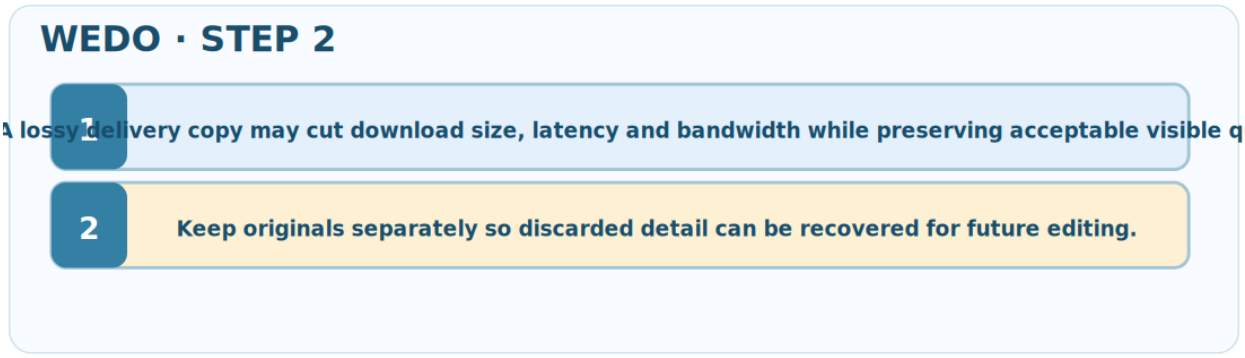
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A web page uses 4K source photographs but displays small thumbnails. Evaluate lossy compression.

A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.



WEDO · STEP 2: A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.; Keep originals separately so discarded detail can be recovered for future editing.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Keep originals separately so discarded detail can be recovered for future editing.

IMPROVE: Explain one correction to your own reasoning.